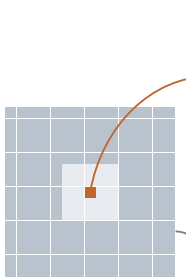
training samples (planted on the GPU in real EnMAP / PRISMA / AVIRIS-NG tiles): $\mathbf{x} = g(\lambda)\left(a\,\mathbf{t} + (1-a)\,\mathbf{b}\right) + \mathbf{n}$, $\mathbf{r}' = g(\lambda)\,\mathbf{r}$, $a \sim$ log-uniform$[0.05, 1]$, g smooth, $\pm 10\%$, $\mathbf{n}$ from the tile's own neighbour differences



scene tile; ring = valid pixels at Chebyshev distance 3–7 (200)

pixel $\mathbf{x} \in \mathbb{R}^N$

ring mean $\mathbf{r} \in \mathbb{R}^N$

library spectrum $\mathbf{t}_j$, $j = 1..L$

family library
USGS / ECOSTRESS
$L = 378$ (EnMAP)

wavelength-aware spectral stem

shared θ_s

wavelength-aware spectral stem

shared θ_s

wavelength-aware spectral stem

shared stem (init. Indradhanu V2)
normalise by $\mu(\lambda), \sigma(\lambda)$ → 40 nm-bin cross-attention → BN (frozen)

pixel tower

64 $\mathbf{z}_x$

64 $\mathbf{z}_r$

$[\,\mathbf{z}_x,\ \mathbf{z}_r,\ \mathbf{z}_x - \mathbf{z}_r\,]$ 192

pixel MLP
192→256→256, GELU

$\mathbf{h}$

Linear → ℓ_2 → $\hat{\mathbf{z}} \in \mathbb{S}^{127}$

Linear → σ → fill $\hat{a} \in [0, 1]$

$0.5\,|\hat{a} - a|$

64 $\mathbf{z}_t$

library MLP
64→256→256→128, GELU

ℓ_2 → $\mathbf{e}_j \in \mathbb{S}^{127}$

library tower

encoded once per band list · a new library is encoded, not retrained

$P(c) = P(\text{man-made}) \cdot P(c \mid \text{man-made})$
$c \in$ {oil, plastic, textile, wood/paper, paint, metal, concrete/asphalt}

class matching

$s_j = \cos(\hat{\mathbf{z}}, \mathbf{e}_j)\,/\,\tau$ — τ learned

$S_c = \log \sum_{j \in c,\ j \neq j^\star} e^{s_j}$ — class pooling

S_{nat}: the same pooling over the *natural* entries — natural = one more class

$P(\text{man-made}) = \sigma\left(\log \sum_c e^{S_c} - S_{\text{nat}} + b\right)$ — level 1

$P(c \mid \text{man-made}) = \text{softmax}_c\, S_c$ — level 2

$\mathcal{L}_{\text{BCE}}$ on $P(\text{man-made})$

$\mathcal{L}_{\text{CE}}$ on $P(c \mid \text{man-made})$, man-made samples;
$j^\star$ = the planted spectrum, left out (train and test)